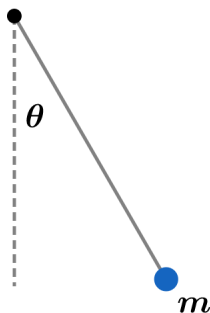


2019B F=ma Exam: Problem 19

Kevin S. Huang



We can check limiting cases:

At $\theta = \pi/2$, the mass is at rest so $T = 0$ and $a = g$.

At $\theta = 0$, the mass has velocity

$$\frac{1}{2}mv^2 = mgR$$

$$v = \sqrt{2gR}$$

so

$$a = a_c = \frac{v^2}{R} = 2g$$

so the answer is A.